

Relevant Reading

- [Computational Aeronautics](#)
 - Chapter 6: Propellers and Turbines (23 pages)
NOTE: Read 6.3 OR 6.4 depending on your application.

Problem 1 Vocabulary

Explain the following terms; making sure to use sufficient detail, including any math or helpful figures. In some cases, these terms are simple one sentence definitions, in others, you should include several paragraphs to explain them fully.

Rotor Analysis

- Blade Element Momentum Theory
 - Conservation of Linear Momentum
 - Conservation of Angular Momentum
 - Blade Element
 - Inflow Angle vs angle of attack vs twist vs pitch
- Blade Loading
- Root Bending Moment

Rotor Performance

- Thrust Coefficient
- Power Coefficient
- Torque Coefficient
- Rotor Efficiency

Problem 2 Theory

- List the assumptions of BEMT. How do they affect the accuracy of the method.
- What are the main limitations of BEMT? Is there anything that they are dependent on that determines the accuracy of the method? Are there any cases where BEMT is not applicable?
- Describe the physical meaning of the axial and tangential induction factors. How do they influence the local flow around a blade element?
- Why is it necessary to couple blade element theory with momentum theory in BEMT? What would be missing if you only used one of the two?

Problem 3 Exploration

Complete the following exploration.

- Find experimental data for a rotor of your choice. This could be a propeller, wind turbine, or helicopter rotor. If the data is not in a form that can be used directly, you may need to digitize it using a tool like [WebPlot-Digitizer](#).
- Implement the rotor in CCBlade.jl and compare the results to the experimental data.
- Determine the error between the experimental data and the CCBlade.jl results. What might cause the source of this error? How could you improve the accuracy of your results?